Reconstructing interior transmission tomographic images with an offset-detector using a deep-neural-network

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ABSTRACT

Interior tomography that acquires truncated data of a specific interior region-of-interest (ROI) is an attractive option to low-dose imaging. However, image reconstruction from such measurement does not yield an accurate solution because of data insufficiency. There have been developed a host of approaches to getting an approximate useful solution including various weighting methods, iterative reconstruction methods, and methods with prior knowledge. In this study, we use a deep-neural-network, which has shown its potentials in various fields including medical imaging, to reconstruct interior tomographic images. We assumed an offset-detector geometry which has wide applications in cone-beam CT (CBCT) imaging for its extended field-of-view (FOV) in this work. We trained a network to synthesize 'ramp-filtered' data within the detector active area so that the corresponding ROI reconstruction would be truncation-artifact-free in the filtered-backprojection (FBP) reconstruction framework. We have compared the results with post- and pre-convolution weighting methods and shown outperformance of the neural network approach.

Keywords: Interior tomography, Truncation artifact, Deep-neural-network.

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1 INTRODUCTION

The focus of this work is on handling transversely truncated data in CBCT image reconstruction. One can effectively increase the size of FOV in cone-beam CT by placing the flat panel detector in an offset geometry, and it constitutes a standard scan protocol in many commercialized clinical CBCT systems. Even with such an extended FOV, the object support size may be larger than that. As a low-dose scanning option, interior tomography that takes projections of an interior ROI only is also under active research. In such cases, transverse data truncation occurs and can cause image artifacts losing image accuracy in the FBP reconstruction framework. A number of approaches to producing an approximate but useful solution have been developed[1]. To name a few, applying weighting functions[2,3], frequency split scheme[4], and iterative methods are available[5].

Recently, deep learning approaches have gained great interest in various fields including medical imaging area. Data-driven nature combined with a series of convolutional operations and rectification allow a trainable network that can do a number of tasks such as classifying objects, filling the missing portions in signals, converting the signals into a different domain. Outperformance of deep learning approaches has also been being actively reported in medicine not only in the analysis[6,7] but also in the reconstruction of medical images[8-11]. The deep learning has also been applied to reconstructing interior tomography in a parallel-beam geometry[12]. In this study, we show that a deep-neural-network (DNN) can provide ramp-filtered data that are truncation-artifact-free for reconstructing interior tomography in CBCT. FBP reconstruction framework is considered throughout the study, and the offset detector geometry has been assumed. Although it does not have to be an offset geometry, we have chosen it for its wider applications in medicine than full-beam geometry.

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2 MATERIALS AND METHODS

We trained a DNN to remove artifact component that is associated with data truncation in the ramp-filtered projection data. Instead of making up synthetic projection data outside the truncated ROI, which is often the case of existing correction method, we would like to convert the ramp-filtered data that contains truncation artifacts to the one that is truncation artifact free. To train the DNN, we collected 45 Lung CT patient data from The Cancer Imaging Archive (TCIA)[13] and forward-projected the images using distance-driven algorithm.[14] Scanning condition that includes an offset-detector and truncation is summarized in Table 1. An example of truncated projection data is shown in Fig. 1 (a). The blank region in the left of the figure is due to the offset-detector geometry.

Table 1. Geometric parameters for simulation

Parameters	Values
Source-to-axis distance	1000mm
Source-to-detector distance	1500mm
Detector height/width	109.375mm/300mm
Detector pixel height/width	0.78125mm/0.78125mm
Detector offset	116 pixels/90.625mm
Number of projection data	360 views
Angle coverage	360˚

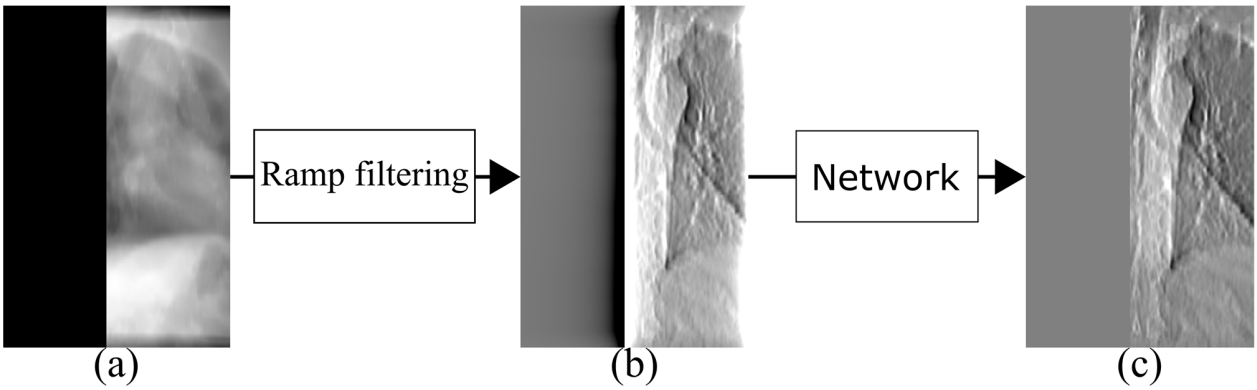


Figure 1. Workflow and data used in this study. (a) Truncated projection data, (b) ramp-filtering result of the truncated projection data, and (c) truncated ramp-filtering result of the non-truncated projection data

Due to data truncation, ramp-filtering invokes inaccurate filtering results particularly including enhanced pixel values near truncated boundaries of the data as shown in Fig. 1 (b). We trained the network to convert the artifact-contaminated filtered data to the artifact-free one as shown in Fig. 1 (c). The artifact-free filtering results were provided in the training period by use of the filtered results of non-truncated projection data that have been separately prepared. We provided projection data (Fig.1 (a)) and corresponding filtered projection data (Fig. 1 (b)) as inputs and artifact-free filtered results (Fig.1 (c)) as desired outputs to the network. We did not reduce the size of the data nor make patches.

We constructed a network in a similar structure to U-Net[15] with replacing pooling layers into strided-convolution as shown in Fig. 2. We used Adam optimizer with learning 0.0001 to minimize the mean-squared error between the output of the network and ground truth data. We had 14580 projection data from training and trained the network for 200 epochs with 3 data per batch. We used the gradient clipping method to make a norm of gradients of each kernel to less than or equal to 1×10^{-5} and the residual learning scheme to make the network converge stably.

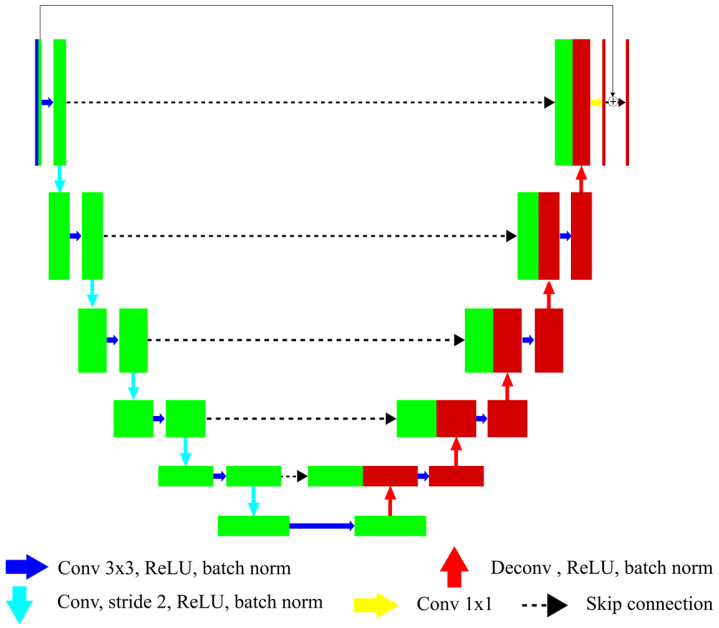


Figure 2. Network structure

We implemented the pre- and post-convolution weighting methods[3] with a cosine extrapolation[16] to complete projection data and compared the results with the proposed ones. We used cosine weighting[17] to normalize redundancy of measurement while reconstructing the results from the neural network.

3 RESULTS

The training and validation loss of the network are shown in Fig. 3.

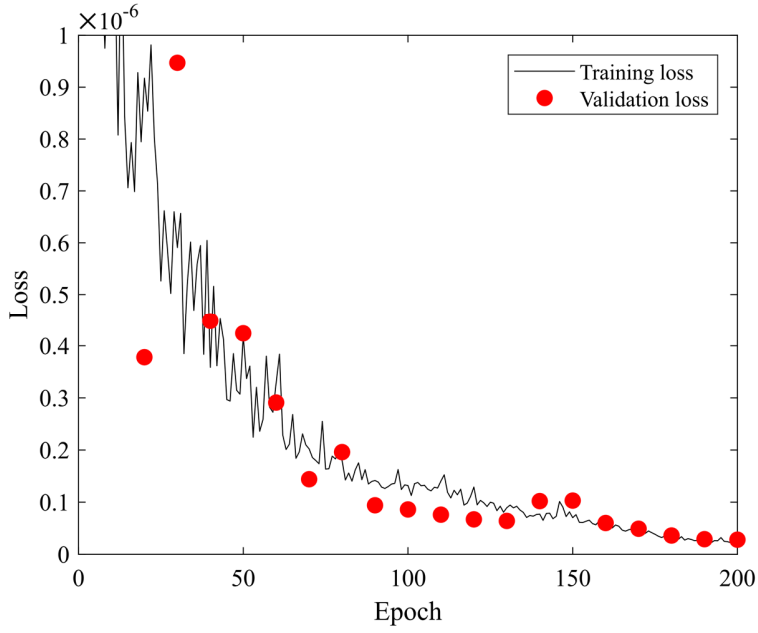


Figure 3. Training and validation loss of the network

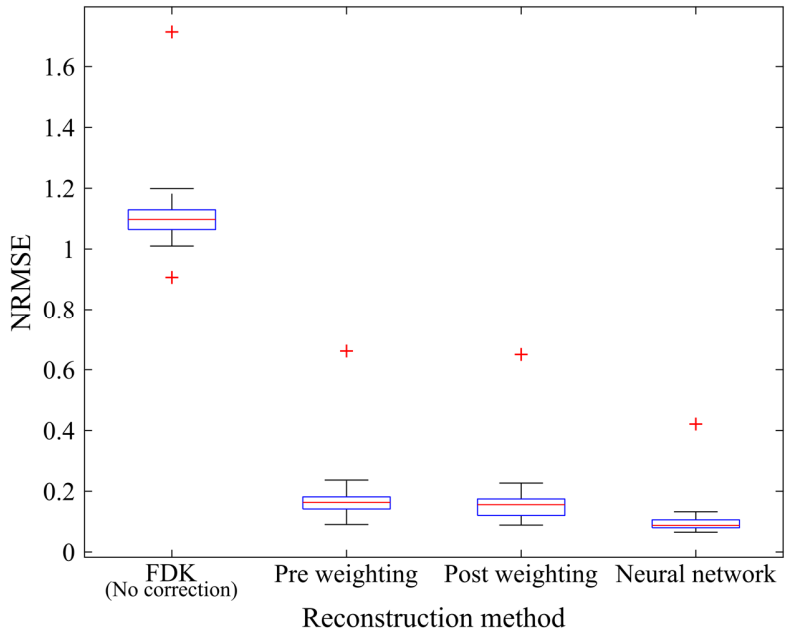


Figure 4. NRMSE comparison results

Another 13 patients' Lung CT data, which were not included in the training phase have been used to evaluate the performance of the network. We compared the results in the reconstructed image domain. In the comparison, we used the reconstructed images without truncation as ground truth data. Normalized root-mean-square error of the reconstructed images using FDK without any correction[18], pre-convolution weighting, post-convolution weighting, and neural network to the ground truth are plotted in Fig. 4. As shown in Fig. 4, results from the neural network have smaller errors than other methods. An example set of the reconstructed images is presented in Fig. 5.

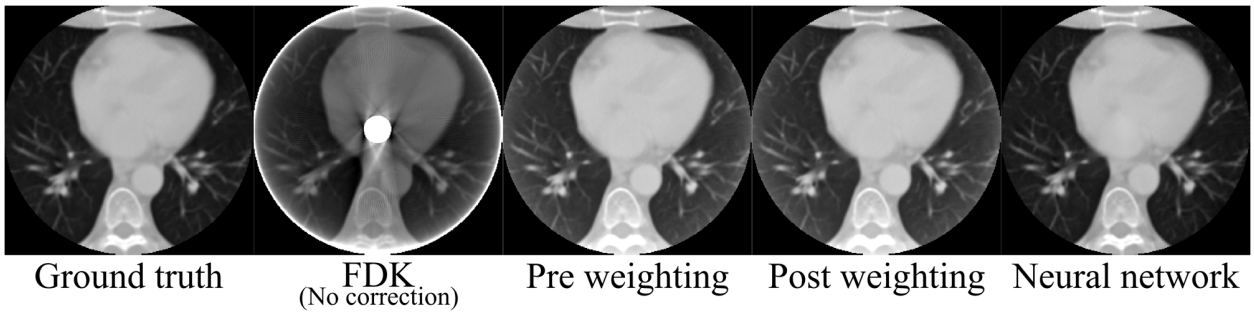


Figure 5. Reconstructed images from truncated projection data with offset-detector, window: [0, 0.034]

4 CONCLUSION

We proposed a deep-neural-network approach for reconstructing interior tomography with an offset-detector geometry. The network was trained to correctly filter the projection data that is truncation artifact free. Reconstructed images by the neural network showed smaller errors than the pre- and post-convolution weighting methods.

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